

EduGenie: Google Gemini Powered Learning Assistant

5. Project Development Phase

Purpose: Implement the EduGenie backend modules, APIs, and web interface.

Phase Activities

- Explanation module uses LaMini-Flan-T5 for concise, context-aware explanations.
- Q&A; module uses Gemini 1.5 Pro for general knowledge and academic questions.
- Quiz module generates three MCQs with four options each and structures the output as JSON.
- Summary module converts long paragraphs into concise versions while retaining core information.
- Learning Path module creates beginner-to-advanced learning recommendations with useful resources.
- FastAPI endpoints are implemented and connected to module logic.
- The frontend contains a task dropdown, text area, submit button, responsive styling, and a result container.
- Form submissions send POST requests to the backend and display results.

Project Focus: This phase is organized from the supplied EduGenie project document while retaining the requested phase name.